1 Temporal Routing and Reasoning-Class Dynamics

We extend aggregate routing statistics with diagnostics of allocation persistence, changes in mixture weights, and the organization of reasoning episodes. The definitions below separate published foundations from adaptations introduced for this study. They measure observed behavior; a large or small value does not, by itself, identify useful reasoning or a beneficial intervention.

1.1 Notation and observation scope

At routed layer ℓ and reasoning-token position t, let $S_{\ell t} \subseteq \{1, \dots, E_\ell\}$ be the native selected expert set, with $|S_{\ell t}| = k_\ell$. Let $p_{\ell t}$ denote the full router probability vector and $a_{\ell te} \geq 0$ the weight actually assigned to a selected expert. The normalized executed mixture is

$$m_{\ell te} = \frac{\mathbf{1}\{e \in S_{\ell t}\}\, a_{\ell te}}{\sum_{j \in S_{\ell t}} a_{\ell tj}}. \tag{1}$$

Thus $p_{\ell t}$ describes router preferences, whereas $m_{\ell t}$ describes the relative weights of the selected routed experts. Neither is reconstructed from scalar confidence summaries. Shared experts are excluded from $S_{\ell t}$ and these routed-expert distributions.

We use a trailing window $W_t^{(w)} = \{t - w + 1, \dots, t\}$ contained in one contiguous reasoning segment. The current configuration uses $w \in \{64, 256\}$, stride 64, and lags $h \in \{1, 4, 16, 64\}$. Prompt tokens, padding and gaps are excluded; insufficient history yields an unavailable metric. All logarithms are natural, so entropies and divergences are in nats. Layer indices are preserved throughout.

1.2 Expert-set persistence and turnover

Let $V_{t,h}^{(w)}$ contain positions $u \in W_t^{(w)}$ for which $u - h$ is observed in the same reasoning segment. Lagged overlap is

$$O_{\ell,h}(t; w) = \frac{1}{|V_{t,h}^{(w)}|} \sum_{u \in V_{t,h}^{(w)}} \frac{|S_{\ell u} \cap S_{\ell,u-h}|}{k_\ell}, \qquad U_\ell(t; w) = 1 - O_{\ell,1}(t; w). \tag{2}$$

Overlap lies in $[0, 1]$ and scores persistence of selected identities; turnover U scores adjacent replacement. Rank changes inside an unchanged set do not increase turnover. The temporal lag construction extends the existing adjacent overlap metric.

Uniform-chance correction. Following the normalized overlap of Olson et al. [5], but applying it to temporal pairs, define

$$O_{\ell,h}^{\text{unif}} = \frac{O_{\ell,h} - k_\ell/E_\ell}{1 - k_\ell/E_\ell}. \tag{3}$$

Zero denotes the uniform independent-selection baseline; one denotes identical sets. Negative values are possible, and the expression is undefined when $k_\ell = E_\ell$. This affine transform is not independent evidence from raw overlap.

Usage-preserving order control. Our additional control permutes routing sets within the observed trailing $w + h$-token pool, clipped at the segment start. If the pool contains n tokens and

expert e appears at n_e positions, the exact expected overlap under sampling distinct token positions is

$$B_{\ell,h} = \frac{\sum_e n_e(n_e-1)}{n(n-1)k_\ell}, \qquad O^{\text{excess}}_{\ell,h} = O_{\ell,h} - B_{\ell,h}. \tag{4}$$

This separates temporal persistence from high overlap caused simply by repeatedly using a small expert pool. Positive excess indicates more persistence than the same usage frequencies predict under random ordering.

1.3 Allocation concentration, diversity and drift

Local and marginal entropy. LIMoE distinguishes token-local entropy from entropy of the average routing distribution [4]. We adapt this distinction from modality-level training regularization to descriptive trailing windows. For $r \in \{p, m\}$, set $\bar{r}_{\ell,W} = w^{-1}\sum_{u\in W} r_{\ell u}$ and

$$\begin{aligned} H^r_{\text{local}}(W) &= \frac{1}{w}\sum_{u\in W} H(r_{\ell u}), & H^r_{\text{marginal}}(W) &= H(\bar{r}_{\ell,W}), \\ H(v) &= -\sum_e v_e \log v_e, & I^r_{\text{alloc}}(W) &= H^r_{\text{marginal}}(W) - H^r_{\text{local}}(W). \end{aligned} \tag{5}$$

Local entropy scores within-token dispersion. Marginal entropy scores diversity across the window. Their nonnegative difference scores how much distributions vary across positions: it is the mutual information between a uniformly sampled window position and an expert drawn from that position's distribution. It is invariant to token order, so it does not itself measure temporal drift.

For native selection counts $c_e = \sum_{u\in W} \mathbf{1}\{e \in S_{\ell u}\}$, we additionally report $H_{\text{selection}} = H((c_e/(wk_\ell))_e)$. This describes usage diversity without using mixture weights. The corresponding selection rates c_e/w sum to k_ℓ, not one.

Effective expert count. We express marginal executed-mixture entropy as

$$N_{\text{eff},\ell}(W) = \exp\big(H^m_{\text{marginal}}(W)\big). \tag{6}$$

It is the number of equally weighted experts with the same window-level entropy, between 1 and E_ℓ. It can exceed k_ℓ because different tokens select different experts; it is not the number executed per token.

Within-window distribution drift. Using Jensen–Shannon divergence [3], split W into equal chronological halves W^- and W^+ and define

$$\begin{aligned} D^r_\ell(W) &= \text{JSD}(\bar{r}_{\ell,W^-}, \bar{r}_{\ell,W^+}), \\ \text{JSD}(a,b) &= H\left(\frac{a+b}{2}\right) - \frac{H(a)+H(b)}{2}. \end{aligned} \tag{7}$$

The score lies in $[0, \log 2]$ and measures redistribution between the two halves. Full-router and executed-mixture drift are reported separately. Our temporal control averages the same statistic over $B = 16$ random permutations of positions inside W, giving $D^{r,\text{shuffle}}$ and $D^{r,\text{excess}} = D^r - D^{r,\text{shuffle}}$. Excess drift can be negative. All permutations use only the observed window.

1.4 Native selection-boundary diagnostics

Let $s_{\ell te}$ be the scores actually used for deterministic top-k selection, including any relevant selection bias. We extend the existing boundary-margin diagnostic with a temporal summary:

$$g_{\ell t} = \min_{e \in S_{\ell t}} s_{\ell te} - \max_{e \notin S_{\ell t}} s_{\ell te}, \qquad \bar{g}_\ell(W) = \frac{1}{w} \sum_{u \in W} g_{\ell u}. \tag{8}$$

A small gap indicates a nearly tied selected/unselected boundary in score space; it does not measure answer robustness. Ties give zero. This metric requires $k_\ell < E_\ell$ and verified native score semantics; unsupported routing rules are marked unavailable.

The study-specific weak-boundary fraction is

$$F_\ell^{\text{weak}}(W) = \frac{1}{w} \sum_{u \in W} \mathbf{1}\{g_{\ell u} \le \theta_\ell\}, \qquad \theta_\ell = Q_{0.10}(g_\ell; \mathcal{D}_{\text{train}}). \tag{9}$$

The threshold is fitted only on calibration data or the training fold, counting each token once despite overlapping windows. A larger fraction means more near-boundary selections relative to that reference. Until such a threshold is fitted, the fraction remains unavailable; the compact correlation report uses the mean gap only.

1.5 Reasoning-class dynamics

ThinkARM motivates describing reasoning through episode sequences and transition patterns [2]. We retain the project's seven labels $\mathcal{C} = \{\text{Read}, \text{Analyze}, \text{Plan}, \text{Implement}, \text{Explore}, \text{Verify}, \text{Monitor}\}$ rather than ThinkARM's eight-class taxonomy. Our censoring and hazard diagnostics are study-specific extensions, not claimed reproductions of ThinkARM metrics. Let C_i be the label of reasoning sentence i and n_i its aligned token count. Unknown labels break sequences rather than becoming an eighth class.

Occupancy, transitions and transition uncertainty. Sentence and token shares are respectively $f_c = \sum_i \mathbf{1}\{C_i = c\}/N$ and $f_c^{\text{tok}} = \sum_i n_i \mathbf{1}\{C_i = c\} / \sum_i n_i$, with denominators restricted to labeled observations and label coverage reported separately. For adjacent labeled sentences in one contiguous segment, let N_{cd} count transitions from c to d. Then

$$P_{cd} = \frac{N_{cd}}{\sum_{b \in \mathcal{C}} N_{cb}}, \qquad P_{\text{stay}}(c) = P_{cc}, \qquad P_{\text{exit}}(c) = 1 - P_{cc}, \qquad H_{\text{next}}(c) = -\sum_d P_{cd} \log P_{cd}. \tag{10}$$

These score destination preference, persistence, switching, and uncertainty over observed next classes. Zero source support gives an undefined rate. Natural termination is outside this seven-class matrix; hence $1 - P_{cc}$ is conditional on an observed next labeled sentence, not an unconditional stopping probability. Conditioning additionally on C_{i-1} describes second-order transition patterns.

Runs, returns and motifs. A run is a maximal contiguous sequence of the same class. Its sentence/token length scores sustained allocation to that reasoning activity; its current age uses only the observed prefix. Entry after a missing-label gap is unobserved, so full duration is unavailable. A gap, cap or unknown ending right-censors the run; observed natural termination is a competing exit event. Return intervals measure the sentence/token distance between successive run starts of the same class, without crossing gaps. Counts of contiguous label bigrams and trigrams describe recurring local motifs. Motif counts alone are not significance scores and do not establish purposeful self-correction.

Destination-specific exit hazards. For runs with observed entry, let $Y_c(a)$ count runs whose outgoing boundary at sentence age a is observable, and $D_{cd}(a)$ count exits to destination d at that age. Our discrete hazard is

$$h_{c\to d}(a) = D_{cd}(a)/Y_c(a), \qquad d \in (\mathcal{C} \setminus \{c\}) \cup \{\text{natural termination}\}. \tag{11}$$

It scores the chance of a particular observed exit at that age, conditional on reaching an observable boundary. Right-censored runs contribute earlier boundaries but not their unobserved final boundary. Empty risk sets remain undefined; observed run lengths must not be mistaken for uncensored dwell times.

1.6 Relating routing to reasoning classes

Within-question class contrasts. For a scalar window metric z, let $\bar{z}_{q,c}$ average windows ending in class c within question q, and $\bar{z}_{q,\neg c}$ average other labeled endpoints. Our contrast is

$$\Delta_c(z) = \frac{1}{|\mathcal{Q}_c|} \sum_{q\in\mathcal{Q}_c} (\bar{z}_{q,c} - \bar{z}_{q,\neg c}), \tag{12}$$

where $\mathcal{Q}_c$ contains questions with both kinds of window. Layers are averaged equally within each question. Positive values indicate larger metric values around that class. A trailing window can include earlier classes, so this is an endpoint-conditioned association. The saved original class-view comparison instead uses class minus all reasoning and is kept as a distinct estimand.

Expert-specific association diagnostics. The broader diagnostic interface also compares expert activation rates, adapting SteerMoE's contrast principle [1] from paired behavioral prompts to observed classes. For tokens with label c, define

$$R_{\ell e|c} = \frac{\sum_t \mathbf{1}\{C(t) = c,\ e \in S_{\ell t}\}}{\sum_t \mathbf{1}\{C(t) = c\}}, \qquad A_{\ell e,c} = R_{\ell e|c} - R_{\ell e|\neg c}. \tag{13}$$

This is a token-weighted selection-rate difference; positive values indicate class-associated recruitment. Expert identity always includes its layer. A secondary normalized pointwise mutual information score, following the association measure used by RICE [6], is

$$\mathrm{nPMI}(X_{\ell e}, c) = \frac{\log[P(X_{\ell e} = 1, C = c)/(P(X_{\ell e} = 1)P(C = c))]}{-\log P(X_{\ell e} = 1, C = c)}, \quad X_{\ell e}(t) = \mathbf{1}\{e \in S_{\ell t}\}. \tag{14}$$

Zero indicates independence; the sign indicates positive or negative association. Degenerate joint probabilities are marked unavailable. Replacing RICE's marker tokens with sentence-class labels is our adaptation, not its published signed marker-ranking score. Contrasts require at least ten supporting questions in each arm and ten with expert activation for inference. These expert-level diagnostics and motif mining are defined by the broader interface but were not part of the compact recovery analysis.

Interpretation and aggregation. Missing data are never encoded as zero. Question-level aggregation and question-bootstrap intervals prevent overlapping windows from being counted as independent observations; searched comparisons require multiple-testing correction. Labels supplied by a retrospective judge describe observed reasoning and are not assumed available to an online controller. These metrics supply hypotheses about routing and reasoning dynamics, not causal claims that persistence, instability, or any class-associated expert is helpful or harmful.

References

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